StudyForge AI

Speaker Notes for Akshita and Aasveer

16-slide presentation script

Presenter allocation: Akshita - slides 1, 3, 5, 6, 7, 9, 11, 13, 15. Aasveer - slides 2, 4, 8, 10, 12, 14, 16.

Slide 1: StudyForge AI

SPEAKER: AKSHITA

Good morning everyone. We are presenting StudyForge AI, a personal AI-powered study companion for students. The idea behind our project is simple: students should be able to keep their study material, questions, revision progress, and study planning in one place. Instead of moving between notes, PDF folders, generic AI tools, and task lists, StudyForge AI brings these activities together in one organised workspace. In this presentation, we will explain the problem, our solution, its features, architecture, and future scope. I will now hand over to Aasveer, who will introduce the team.

Slide 2: Team and roles

SPEAKER: AASVEER

This slide introduces our project team and the members who worked on StudyForge AI. Our work involved planning the problem statement, designing the user experience, building the frontend and backend, and testing the system. Even though different members contributed to different parts, the project was developed as one integrated platform. We will now explain why such a platform is useful for students. I hand over to Akshita.

Slide 3: Why students need StudyForge AI

SPEAKER: AKSHITA

Students often keep notes, PDFs, assignments, and links across several applications and devices. This makes it difficult to find the right material when revision time is limited. Another challenge is understanding difficult concepts from personal study material, because generic answers may not match the notes a student is using. Students also need help with revision planning, deadlines, and viva preparation. StudyForge AI addresses these problems through a single platform where students can upload material, ask questions, monitor revision, plan tasks, and continue saved conversations. Aasveer will now describe these problems in more detail.

Slide 4: Problem statement

SPEAKER: AASVEER

We identified three main challenges. First is organisation: notes and PDFs become scattered across devices and folders. Second is understanding: complex topics take time to learn, while normal AI tools may answer without using the student's own study material. Third is preparation: students may not know which topic to revise first, may forget deadlines, and can feel unprepared for a viva. These challenges show the need for one student-friendly system that combines material management with context-aware study support. Akshita will now present our proposed solution.

Slide 5: StudyForge AI capabilities

SPEAKER: AKSHITA

StudyForge AI works as a web-based personal study assistant. The first capability is upload and organise: students can create subjects, upload PDFs, and access their material from one central place. The second capability is AI assistance: students can ask questions about selected documents and choose a study mode such as Explain Simply, Debug Code, Practise Viva, or Solve Step by Step. The third capability is tracking and planning: students can record revision confidence, manage tasks and deadlines, and save chats for later. Next, Aasveer will explain the objectives behind these features.

Slide 6: Objectives

SPEAKER: AASVEER

Our first objective is to centralise academic material so students can store and find PDFs, notes, and study resources easily. The second objective is to support different learning needs through AI study modes. A student may need a simple explanation, help debugging code, a viva question, or a step-by-step solution. Our third objective is revision and self-assessment. The platform lets students track their confidence in topics and keep useful conversations for assignments, code doubts, and exam preparation. Akshita will now show the main features of the platform.

Slide 7: Main features

SPEAKER: AKSHITA

The first main feature is the Subject Library. Students can create subjects, upload PDF material, and later search, filter, or download their notes from one location. The second feature is the AI Study Assistant. After choosing a document, the student can ask a question and select a suitable mode, including Explain Simply, Give a Hint, Debug My Code, Practise Viva, Solve Step by Step, or Plan My Study. These features make the platform useful for different types of academic work. Aasveer will now explain the revision tracker and planner.

Slide 8: Revision tracker and planner

SPEAKER: AASVEER

The Revision Tracker helps students add topics and mark their current status as Learning, Needs Revision, or Confident. This makes weak areas visible before an exam. The My Planner feature lets students add, edit, or delete tasks such as assignments, exams, revision sessions, and doubts. They can also set optional due dates. Together, these features support better time management because students can see both their revision progress and the tasks they still need to complete. Akshita will now walk through the system workflow.

Slide 9: Working of the system

SPEAKER: AKSHITA

The workflow begins when a student signs in, selects or creates a subject, and uploads or chooses a PDF from the subject library. Next, the student selects a study mode and asks a question related to the material. The system sends the question, selected document, study mode, and relevant context to the AI service. The student then receives an answer and can save the conversation for later use. This gives students a guided flow from organising material to receiving study support. Aasveer will now explain the architecture behind this workflow.

Slide 10: System architecture

SPEAKER: AASVEER

StudyForge AI uses a layered architecture. On the frontend, React and Vite create a responsive interface for the dashboard, library, AI chat, revision tracker, and planner. The backend uses FastAPI for authentication, PDF processing, chat management, revision records, and planning tasks. SQLite stores user profiles, subjects, tasks, and conversations. Gemini AI generates intelligent responses, while PyMuPDF extracts text from uploaded PDFs. This separation keeps the user interface, application logic, storage, and AI processing organised. Akshita will now cover the technology stack.

Slide 11: Technology stack

SPEAKER: AKSHITA

For the frontend, we use React for component-based interface development and Vite for fast development and hot module replacement. HTML, CSS, and JavaScript support the styling and interactions. For the backend, we use Python with FastAPI, and SQLite stores user profiles, subjects, revision records, tasks, and chat history. For the AI and document layer, Gemini provides responses and PyMuPDF extracts PDF text. We also use Pytest and Playwright to test backend functions and user workflows. Aasveer will now show the project visualisation.

Slide 12: Project visualisation

SPEAKER: AASVEER

This slide presents the visual side of our project. The purpose of the interface is to make study work feel organised instead of overwhelming. The dashboard brings together the key areas: subjects and study material, AI assistance, revision progress, and planning. A student can move from a document to a question, then save the answer and return to it later. The visual design supports the same goal as the technical design: one clear workspace for everyday learning and exam preparation. Akshita will now describe how we tested the system.

Slide 13: Testing and validation

SPEAKER: AKSHITA

We validated both the backend and frontend. On the backend, Pytest checks API endpoints and data handling, including authentication, subject management, PDF processing, and saved chats. On the frontend, we verified the production build and tested user navigation across the dashboard, library, chat, revision tracker, and planner. We also checked important functions such as subject and document management, revision tracking, saved conversations, code copying, and responsive mobile layout. These checks help confirm that the application works as an integrated study platform. Aasveer will now explain the current limitations.

Slide 14: Limitations

SPEAKER: AASVEER

Like any AI-based system, StudyForge AI has limitations. Its AI features depend on Gemini availability and quota limits, so response time may vary and the service can be temporarily unavailable. PDF handling also has limits: large files and low-quality scanned documents can be difficult to process. The current upload limit is approximately 20 MB and 200 pages. Finally, the planner does not yet send automatic reminders, and revision confidence is entered by the student rather than calculated automatically. Akshita will now present the future scope.

Slide 15: Future scope

SPEAKER: AKSHITA

There are several planned improvements. For automation, the platform could provide reminders, notifications, personalised study recommendations, quizzes, and performance analytics. For accessibility, we can add multilingual support, voice interaction, better accessibility features, and improved handling of handwritten or scanned notes. For collaboration, cloud storage could support multi-device access, group study spaces, and synchronisation across platforms. These additions would make StudyForge AI more useful for a wider range of students and study situations. Aasveer will now conclude the presentation.

Slide 16: Conclusion

SPEAKER: AASVEER

To conclude, StudyForge AI combines document-based learning, AI assistance, revision tracking, task planning, and saved conversations in one personalised study platform. The project focuses on helping students organise their work, understand material in context, and prepare more confidently for assignments, exams, and viva sessions. We have also identified realistic limitations and future improvements, which gives the project a clear direction for further development. Thank you for listening. We welcome your questions.